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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST (SET – A)** | **Time:** 1 Hour
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: 1 Mark Questions (3 MCQs + 2 A–R) MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The distance between the points $(0, 5)$ and $(-5, 0)$ is: [1]

(A) 5 **(B)** $5\sqrt{2}$ **(C)** $2\sqrt{5}$ **(D)** 10

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $5\sqrt{2}$

Given: Let the given points be $A(x_1, y_1) = (0, 5)$ and $B(x_2, y_2) = (-5, 0)$.
We know that the distance between two points (x_1, y_1) and (x_2, y_2) is given by:

Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Substituting the given coordinates step by step:

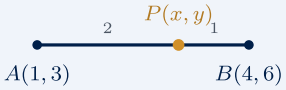
$$AB = \sqrt{(-5-0)^2 + (0-5)^2} = \sqrt{(-5)^2 + (-5)^2} = \sqrt{25+25} = \sqrt{50} = \sqrt{25 \times 2} = \boxed{\mathbf{5\sqrt{2}}\text{ units}}$$

Q2. The coordinates of the point which divides the line segment joining the points $(1, 3)$ and $(4, 6)$ internally in the ratio $2 : 1$ are: [1]

(A) $(2, 4)$ **(B)** $(3, 5)$ **(C)** $(4, 2)$ **(D)** $(5, 3)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $(3, 5)$



Given: $A(x_1, y_1) = (1, 3)$, $B(x_2, y_2) = (4, 6)$, and internal ratio $m_1 : m_2 = 2 : 1$. We apply the internal section formula.

We know that by the internal section formula, the dividing point $P(x, y)$ is:

Formula Used: $P(x, y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$

Substituting $m_1 = 2$, $m_2 = 1$, $x_1 = 1$, $y_1 = 3$, $x_2 = 4$, and $y_2 = 6$:

$$x = \frac{2(4) + 1(1)}{2 + 1} = \frac{8 + 1}{3} = \frac{9}{3} = 3$$
$$y = \frac{2(6) + 1(3)}{2 + 1} = \frac{12 + 3}{3} = \frac{15}{3} = 5$$

$$\boxed{P(x, y) = (\mathbf{3}, \mathbf{5})}$$

Q3. If the midpoint of the line segment joining the points $A(3,4)$ and $B(k,6)$ is $P(x,y)$ and $x+y-10=0$, then the value of k is: [1]

(A) 6 **(B)** 7 **(C)** 8 **(D)** 9

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 7

Given: Endpoints $A(3,4)$ and $B(k,6)$, midpoint $P(x,y)$, and line condition $x+y-10=0$.
We know that the midpoint formula for two points (x_1,y_1) and (x_2,y_2) is:

Formula Used: $P(x,y)=\left(\frac{x_1+x_2}{2},\ \frac{y_1+y_2}{2}\right)$

Computing the coordinates of midpoint $P(x,y)$:

$$x=\frac{3+k}{2},\qquad y=\frac{4+6}{2}=\frac{10}{2}=5$$

Substituting $x=\frac{3+k}{2}$ and $y=5$ into $x+y-10=0$:

$$\frac{3+k}{2}+5-10=0 \implies \frac{3+k}{2}-5=0 \implies \frac{3+k}{2}=5 \implies 3+k=10 \implies \boxed{k=7}$$

Q4. **Assertion (A):** The distance between the points $P(-2,0)$ and $Q(2,0)$ is 4 units.
Reason (R): Mid-point of a line segment divides it in the ratio $1:1$. [1]

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Evaluating Assertion (A): Given $P(-2,0)$ and $Q(2,0)$. We know the distance formula:

Formula Used: $PQ=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Substituting the coordinates:

$$PQ=\sqrt{(2-(-2))^2+(0-0)^2}=\sqrt{(2+2)^2+0}=\sqrt{4^2}=\sqrt{16}=\boxed{4 \text{ units}}$$

Hence, Assertion (A) is **true**.
Evaluating Reason (R): Since a midpoint is equidistant from both endpoints ($m_1=m_2$), it divides any line segment internally in the ratio $1:1$. Hence, Reason (R) is also **true**.
However, the midpoint ratio does not explain the distance formula calculation for PQ. Therefore, both (A) and (R) are true, but (R) is not the correct explanation of (A).

Q5. **Assertion (A):** The point $(0, 2)$ is the point of intersection of the y-axis with the line segment joining $(1, 1)$ and $(-1, 3)$.
Reason (R): The coordinates of a point dividing the line segment joining (x_1, y_1) and (x_2, y_2) in the ratio $m_1 : m_2$ is given by $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$. [1]

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Evaluating Reason (R): The stated section formula is exact and standard, so Reason (R) is **true**.
Evaluating Assertion (A): Let a point $P(0, y)$ on the y-axis divide the segment joining $A(1, 1)$ and $B(-1, 3)$ internally in the ratio $m_1 : m_2 = k : 1$. We apply the section formula from (R):

Formula Used: $P(x, y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$

Equating the x-coordinate to 0 (since P lies on the y-axis):

$$\frac{k(-1) + 1(1)}{k + 1} = 0 \implies -k + 1 = 0 \implies k = 1 \quad \text{(ratio } 1 : 1\text{)}$$

Substituting $k = 1$ to find the y-coordinate:

$$y = \frac{1(3) + 1(1)}{1 + 1} = \frac{4}{2} = 2 \implies \boxed{P = (\mathbf{0}, \mathbf{2})}$$

Thus, both Assertion (A) and Reason (R) are true, and (R) is the correct explanation of (A).

SECTION B: Short Answer Type Questions

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Q6. Find the distance between the points $P(a, b)$ and $Q(-a, -b)$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula Declaration [0.5 Mark]

Given: Points $P(x_1, y_1) = (a, b)$ and $Q(x_2, y_2) = (-a, -b)$.
We know that the distance between (x_1, y_1) and (x_2, y_2) is given by:

Formula Used: $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

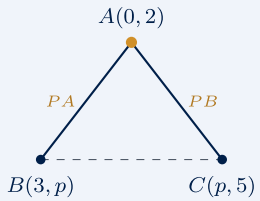
Step 2: Substitution and Algebraic Simplification [1.5 Marks]

Substituting $x_1 = a$, $y_1 = b$, $x_2 = -a$, and $y_2 = -b$:

$$\begin{aligned} PQ &= \sqrt{(-a - a)^2 + (-b - b)^2} \\ &= \sqrt{(-2a)^2 + (-2b)^2} \\ &= \sqrt{4a^2 + 4b^2} = \sqrt{4(a^2 + b^2)} = \boxed{\mathbf{2\sqrt{a^2 + b^2}}\ \text{units}} \end{aligned}$$

Q7. If the point $A(0,2)$ is equidistant from the points $B(3,p)$ and $C(p,5)$, find the value of p. Also find the length of line segment AB. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Step 1: Equating Distances ($AB = AC$) [1 Mark]
Given: $A(0,2)$ is equidistant from $B(3,p)$ and $C(p,5)$, so $AB = AC$, i.e., $AB^2 = AC^2$. We apply the distance formula.

We know that the distance formula between (x_1, y_1) and (x_2, y_2) is:

Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ or $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Applying $AB^2 = AC^2$ and expanding both sides using $(a-b)^2 = a^2 - 2ab + b^2$:

$$(3-0)^2 + (p-2)^2 = (p-0)^2 + (5-2)^2$$

$$9 + (p^2 - 4p + 4) = p^2 + 3^2$$

$$p^2 - 4p + 13 = p^2 + 9$$

Cancelling p^2 from both sides and transposing constants:

$$-4p = 9 - 13 \implies -4p = -4 \implies p = \frac{-4}{-4} = \mathbf{1}$$

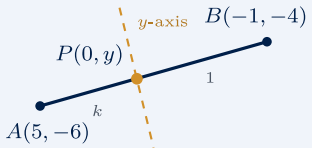
Step 2: Finding Length of AB [1 Mark]
Substituting $p = 1$, we get $B(3,1)$. Using the distance formula for $A(0,2)$ and $B(3,1)$:

$$AB = \sqrt{(3-0)^2 + (1-2)^2} = \sqrt{3^2 + (-1)^2} = \sqrt{9+1} = \sqrt{10} \text{ units}$$

$$\boxed{\mathbf{p = 1} \quad \text{and} \quad \mathbf{AB = \sqrt{10}} \text{ units}}$$

Q8. Find the ratio in which the y-axis divides the line segment joining the points $A(5,-6)$ and $B(-1,-4)$. Also find the coordinates of the point of division. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Step 1: Finding Ratio Using x-Coordinate [1 Mark]
Given: $A(5,-6)$, $B(-1,-4)$. Any point on the y-axis has coordinates $P(0,y)$. Let P divide AB in the ratio $k:1$.

We know the internal section formula:

Formula Used: $P(x,y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$

Substituting $m_1 = k$, $m_2 = 1$, $x_1 = 5$, $x_2 = -1$ and equating the abscissa to 0:

$$\frac{k(-1) + 1(5)}{k+1} = 0 \implies -k + 5 = 0 \implies k = 5 \implies \text{Ratio} = \mathbf{5:1}$$

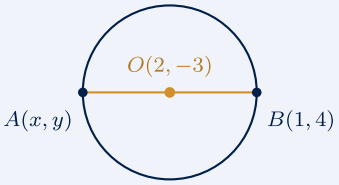
Step 2: Finding Coordinates of the Point of Division [1 Mark]
Substituting $k = 5$, $y_1 = -6$, and $y_2 = -4$ into the ordinate formula:

$$y = \frac{5(-4) + 1(-6)}{5+1} = \frac{-20-6}{6} = \frac{-26}{6} = -\frac{13}{3}$$

$$\boxed{\text{Ratio } = \mathbf{5:1} \quad \text{and} \quad \text{Point of division } = \left(\mathbf{0}, -\frac{\mathbf{13}}{\mathbf{3}}\right)}$$

Q9. Find the coordinates of a point A, where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$. **[2]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Step 1: Setup Using Centre as Midpoint [0.5 Mark]
Given: Centre $O(2, -3)$ and endpoint $B(1, 4)$. Let point A be (x, y). Since centre O is the midpoint of diameter AB, we apply the midpoint formula.

We know the midpoint formula:

Formula Used: $O = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Step 2: Equating Coordinates to Solve for (x, y) [1.5 Marks]
Equating the coordinates of centre $O(2, -3)$ to the midpoint of $A(x, y)$ and $B(1, 4)$:

$$\left(\frac{x+1}{2}, \frac{y+4}{2}\right) = (2, -3)$$

Comparing the x-coordinates and y-coordinates separately:

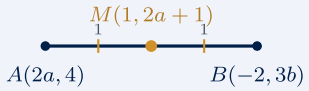
$$\frac{x+1}{2} = 2 \implies x + 1 = 4 \implies x = 4 - 1 = 3$$

$$\frac{y+4}{2} = -3 \implies y + 4 = -6 \implies y = -6 - 4 = -10$$

Coordinates of $A = \mathbf{(3, -10)}$

Q10. If the midpoint of the line segment joining $A(2a, 4)$ and $B(-2, 3b)$ is $M(1, 2a + 1)$, find the numerical values of a and b. **[2]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Step 1: Midpoint Formula Declaration [0.5 Mark]
Given: $A(2a, 4)$, $B(-2, 3b)$, and midpoint $M(1, 2a + 1)$.

We know that the midpoint of (x_1, y_1) and (x_2, y_2) is:

Formula Used: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Step 2: Solving for a and b [1.5 Marks]
Substituting $A(2a, 4)$ and $B(-2, 3b)$ and equating to $M(1, 2a + 1)$:

$$\left(\frac{2a + (-2)}{2}, \frac{4 + 3b}{2}\right) = (1,\ 2a + 1)$$

Equating the abscissae to find a:

$$\frac{2a - 2}{2} = 1 \implies a - 1 = 1 \implies a = 2$$

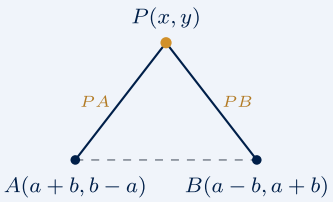
Substituting $a = 2$ and equating the ordinates to find b:

$$\frac{4 + 3b}{2} = 2(2) + 1 \implies \frac{4 + 3b}{2} = 5 \implies 4 + 3b = 10 \implies 3b = 6 \implies b = 2$$

$\mathbf{a = 2}$ and $\mathbf{b = 2}$

Q11. If the point $P(x, y)$ is equidistant from the points $A(a+b, b-a)$ and $B(a-b, a+b)$, prove rigorously that: $bx = ay$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Applying the Equidistant Condition [1 Mark]
Given: $P(x, y)$ is equidistant from $A(a+b, b-a)$ and $B(a-b, a+b)$, so $PA = PB \implies PA^2 = PB^2$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Writing $PA^2 = PB^2$ explicitly using the distance formula:

$$[x - (a+b)]^2 + [y - (b-a)]^2 = [x - (a-b)]^2 + [y - (a+b)]^2$$

Step 2: Full Binomial Expansion on Both Sides [2 Marks]
Using the algebraic identity $(u - v)^2 = u^2 - 2uv + v^2$ on all four squared terms:

$$\text{LHS} = x^2 - 2(a+b)x + (a+b)^2 + y^2 - 2(b-a)y + (b-a)^2$$
$$\text{RHS} = x^2 - 2(a-b)x + (a-b)^2 + y^2 - 2(a+b)y + (a+b)^2$$

Step 3: Cancellation of Identical Terms and Final Proof [2 Marks]
Notice that x^2, y^2, $(a+b)^2$, and $(b-a)^2 = (a-b)^2$ appear equally on both sides. Cancelling them gives:

$$-2(a+b)x - 2(b-a)y = -2(a-b)x - 2(a+b)y$$

Dividing throughout by -2:

$$(a+b)x + (b-a)y = (a-b)x + (a+b)y$$

Collecting x-terms on the left and y-terms on the right:

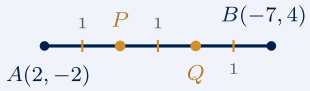
$$(a+b)x - (a-b)x = (a+b)y - (b-a)y$$
$$(a+b-a+b)x = (a+b-b+a)y$$
$$2bx = 2ay$$

Dividing both sides by 2:

$$\boxed{\mathbf{bx = ay} \quad \text{(Hence Proved)}}$$

Q12. Answer the following questions regarding points of trisection:
(i) Define what is meant by the points of trisection of a line segment. [1]
(ii) Find the coordinates of the points of trisection of the line segment joining the points $A(2,-2)$ and $B(-7,4)$. [4] [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



(i) Definition of Points of Trisection [1 Mark]
The **points of trisection** of a line segment AB are the two points P and Q lying on AB that divide it into **three equal parts**, i.e., $AP = PQ = QB$.

We apply the internal section formula:

Formula Used: $P(x,y) = \left(\dfrac{m_1x_2 + m_2x_1}{m_1+m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1+m_2}\right)$

(ii) Finding Coordinates of P and Q [4 Marks]
Given: $A(2,-2)$ and $B(-7,4)$. Since $AP = PQ = QB = 1$ part:

Case 1: Point P divides AB internally in the ratio $1:2$ ($m_1 = 1$, $m_2 = 2$): [2 Marks]

$$P(x_P, y_P) = \left(\frac{1(-7)+2(2)}{1+2}, \frac{1(4)+2(-2)}{1+2}\right) = \left(\frac{-7+4}{3}, \frac{4-4}{3}\right) = \left(\frac{-3}{3}, \frac{0}{3}\right) = (\mathbf{-1, 0})$$

Case 2: Point Q divides AB internally in the ratio $2:1$ ($m_1 = 2$, $m_2 = 1$): [2 Marks]

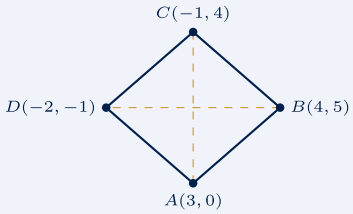
$$Q(x_Q, y_Q) = \left(\frac{2(-7)+1(2)}{2+1}, \frac{2(4)+1(-2)}{2+1}\right) = \left(\frac{-14+2}{3}, \frac{8-2}{3}\right) = \left(\frac{-12}{3}, \frac{6}{3}\right) = (\mathbf{-4, 2})$$

Points of trisection are $(\mathbf{-1, 0})$ and $(\mathbf{-4, 2})$

Q13. The vertices of a rhombus taken in order are $A(3,0)$, $B(4,5)$, $C(-1,4)$ and $D(-2,-1)$.
(i) Calculate the lengths of its diagonals AC and BD using the distance formula. [3]
(ii) Hence, find the area of the rhombus. [Hint: Area of rhombus $= \frac{1}{2} \times d_1 \times d_2$] [2] [5]

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► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



(i) Calculating Lengths of Diagonals AC and BD [3 Marks]
Given: Vertices $A(3,0)$, $B(4,5)$, $C(-1,4)$, $D(-2,-1)$ in cyclic order. We apply the distance formula to find diagonals AC and BD.

We know the distance formula:

Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Length of diagonal AC joining $A(3,0)$ and $C(-1,4)$: [1.5 Marks]

$$d_1 = AC = \sqrt{(-1-3)^2 + (4-0)^2} = \sqrt{(-4)^2 + 4^2} = \sqrt{16+16} = \sqrt{32} = \sqrt{16 \times 2} = \mathbf{4\sqrt{2}} \text{ units}$$

Length of diagonal BD joining $B(4,5)$ and $D(-2,-1)$: [1.5 Marks]

$$d_2 = BD = \sqrt{(-2-4)^2 + (-1-5)^2} = \sqrt{(-6)^2 + (-6)^2} = \sqrt{36+36} = \sqrt{72} = \sqrt{36 \times 2} = \mathbf{6\sqrt{2}} \text{ units}$$

(ii) Calculating Area of Rhombus $ABCD$ [2 Marks]
We know that the area of a rhombus in terms of its diagonals d_1 and d_2 is:

Formula Used: Area of a rhombus $= \frac{1}{2} \times d_1 \times d_2$

Substituting $d_1 = 4\sqrt{2}$ units and $d_2 = 6\sqrt{2}$ units:

$$\text{Area} = \frac{1}{2} \times 4\sqrt{2} \times 6\sqrt{2} = \frac{1}{2} \times (4 \times 6) \times (\sqrt{2} \times \sqrt{2}) = \frac{1}{2} \times 24 \times 2 = \mathbf{24} \text{ sq. units}$$

$$\boxed{AC = \mathbf{4\sqrt{2}} \text{ units}, \quad BD = \mathbf{6\sqrt{2}} \text{ units}, \quad \text{Area} = \mathbf{24} \text{ sq. units}}$$

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